

Evaluating land consolidation decisions for smallholder farmers in Kenya

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Introduction

This document is a supplementary resource describing the land consolidation model used to simulate the outcomes of the decision to consolidate land in Kenya by smallholder farmers. The data are expert estimates collected in March 2024, after co-creating a conceptual model with stakeholders, supplemented with published literature where available and where unavailable other stakeholders not in the workshop were contacted to discuss the estimates.

Required packages

The core R packages are `decisionSupport` (Luedeling et al. 2025), `tidyverse` (Wickham 2023), and `ggplot2` (Wickham 2016).

From conceptual model to a land consolidation function

The conceptual model, co-created with stakeholders, reflects the costs, benefits, and risks that could arise when shifting from small, highly diversified farms to a model where farmers consolidate their land under external management.

Costs a farmer could incur under land consolidation

- Planning costs (legal fees, knowledge acquisition, stationery, printing)
- Loss of natural assets, e.g. perennial plants such as trees
- Food costs no longer saved
- Household upkeep cost prior to the first payment
- Medical bills due to a reduced active lifestyle

Benefits a farmer could gain under land consolidation

- Lease income
- Saved production costs
- Alternative income from on- or off-farm activities
- Better childhood outcomes
- Social cohesion
- Saved medical bills related to land-use accidents/diseases

Costs incurred by a farmer who remains in the current system (“do nothing”)

- Planning costs (incurred whether or not the farmer changes)
- Medical bills related to land-use accidents/diseases and stress
- Production costs

Benefits under the “do nothing” option

- Land security (valued via trees)
- Annual crop yield
- Saved food costs

Both options face risks: farmers who shift face social risks from freed-up time, while those who do nothing face crop-related risks.

Shifting to land consolidation is modeled over a 25-year period, during which the farmer can no longer farm the consolidated parcel but receives lease income commensurate with its size.

The `land_consolidation_function` defines these costs, benefits, and risks for both options. The **decision** is the difference between the two options.

Inputs, model function, and test variables

The input file which contains the variables and estimates -> `land_consolidation_input.csv`. Market scenario file based on value of information -> `market_scenario.csv` which is used later.

```
input_path      <- "land_consolidation_input.csv"
base_estimate  <- decisionSupport::estimate_read_csv(input_path, sep = ";")
input_table    <- read.csv(input_path, sep = ";")
```

```
make_variables <- function(est, n = 1) {
  x <- decisionSupport::random(rho = est, n = n)
  for (i in colnames(x)) assign(i, as.numeric(x[1, i]), envir = .GlobalEnv)
}
make_variables(base_estimate)
```

```
source("land_consolidation_function.R")
```

Monte Carlo simulation

We run a Monte Carlo simulation of the model function using the inputs in `land_consolidation_input.csv`, and in this model we use 10,000 model runs. Full model described here can be found in the `land_consolidation_function.R`

```
mcSimulation_results <- decisionSupport::mcSimulation(
  estimate          = base_estimate,
  model_function    = land_consolidation_function,
  numberOfModelRuns = 1e2,
  functionSyntax    = "plainNames")
```

Results

Distribution of the outcomes

We present 2 key results:- (i) The distribution of the 2 decision options (i) land consolidation and (ii) maintain farms; and (ii) the distribution of decision do which is reflected by the difference of the options.

```
# Assemble model inputs (x) and outputs (y), then express NPVs per hectare.
```

```
mc_output <- cbind(as.data.frame(mcSimulation_results$y),  
                  as.data.frame(mcSimulation_results$x)) %>%  
  mutate(NPV_per_ha = benefit_lc_kes / ha_per_hh,  
         No_NPV_per_ha = benefit_status_quo_kes / ha_per_hh,  
         NPV_per_ha_decision = net_benefit_lc_kes / ha_per_hh)
```

```
# Summary statistics (million KES).
```

```
summarised_stats <- mc_output %>%  
  select(NPV_per_ha, No_NPV_per_ha, NPV_per_ha_decision) %>%  
  summarise(across(everything(), list(  
    min = ~min(., na.rm = TRUE) / 1e6,  
    q05 = ~quantile(., 0.05, na.rm = TRUE) / 1e6,  
    median = ~median(., na.rm = TRUE) / 1e6,  
    q95 = ~quantile(., 0.95, na.rm = TRUE) / 1e6,  
    max = ~max(., na.rm = TRUE) / 1e6,  
    pc_positive = ~mean(. > 0, na.rm = TRUE) * 100))) %>%  
  pivot_longer(everything(),  
               names_to = c("variable", "statistic"),  
               names_pattern = "^(.*)_($)", values_to = "value") %>%  
  pivot_wider(names_from = statistic, values_from = value) %>%  
  mutate(across(min:max, \(x) round(x, 2)))  
summarised_stats
```

```
## # A tibble: 6 x 7
```

##	variable	min	q05	median	q95	max	positive
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 NPV_per_ha	-528.	-33.1	7.89	31.8	90.8	NA
##	2 NPV_per_ha_pc	NA	NA	NA	NA	NA	79
##	3 No_NPV_per_ha	-0.26	2.76	17.1	63.6	726.	NA
##	4 No_NPV_per_ha_pc	NA	NA	NA	NA	NA	99
##	5 NPV_per_ha_decision	-1254.	-89.8	-8.76	17.1	29.7	NA
##	6 NPV_per_ha_decision_pc	NA	NA	NA	NA	NA	33

```
# (a) Overlaid densities of the two options (trimmed to the 5-95% range).
```

```
mc_output %>%  
  select(NPV_per_ha, No_NPV_per_ha) %>%  
  pivot_longer(everything(), names_to = "variables", values_to = "project_outcome") %>%  
  group_by(variables) %>%  
  filter(project_outcome >= quantile(project_outcome, 0.05, na.rm = TRUE),  
         project_outcome <= quantile(project_outcome, 0.95, na.rm = TRUE)) %>%  
  ungroup() %>%  
  mutate(variables = factor(variables, levels = c("NPV_per_ha", "No_NPV_per_ha"),  
                           labels = c("Land consolidation", "Maintain farms"))) %>%  
  ggplot(aes(project_outcome / 1e6, colour = variables, fill = variables)) +  
  geom_density(alpha = 0.3) +
```

```

scale_colour_manual(values = pal_decision) +
scale_fill_manual(values = pal_decision) +
labs(x = "NPV per hectare (million KES)", y = "Density", title = "") + #Overall outcomes
theme_lc +
theme(legend.position = "top")

```

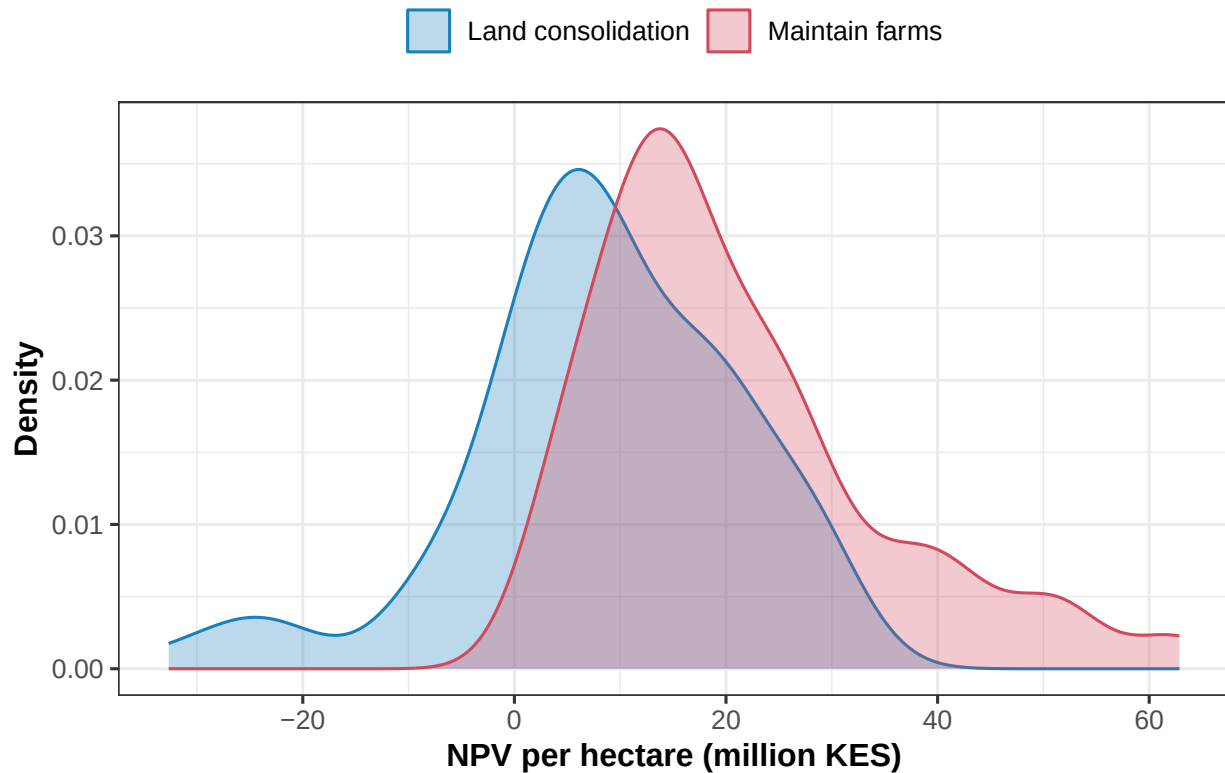


Figure 1: Distribution of the outcomes of land consolidation vs do-nothing options given as net present value per hectare. Based on 10,000 Monte Carlo model runs.

```

# (b) The decision (net benefit), coloured by sign, with directional guidance.
percentages <- mc_output %>%
  summarise(negative = mean(net_benefit_lc_kes < 0, na.rm = TRUE) * 100,
            positive = mean(net_benefit_lc_kes >= 0, na.rm = TRUE) * 100)
negative_percent <- round(percentages$negative, 2)
positive_percent <- round(percentages$positive, 2)

mc_output %>%
  mutate(outcome_category = ifelse(net_benefit_lc_kes >= 0, "positive", "negative")) %>%
  ggplot(aes(net_benefit_lc_kes / 1e6, fill = outcome_category)) +
  geom_histogram(breaks = seq(-90, 90, by = 1)) +
  scale_fill_manual(values = c(negative = "#1a80bb", positive = "#d1495b")) +
  guides(fill = "none") +
  annotate("text", label = paste0(negative_percent, " %"), x = -34, y = 200) +
  annotate("text", label = paste0(positive_percent, " %"), x = 20, y = 200) +

```

```

labs(x = "NPV per hectare (million KES)", y = "Frequency",
     title = "b)") + # Decision outcome for land consolidation
theme_lc +
theme(plot.margin = margin(0, 0, 90, 0)) +
coord_cartesian(clip = "off") +
annotation_custom(linesGrob(arrow = arrow(length = unit(0.25, "cm"), type = "open"),
    gp = gpar(col = "black")),
    xmin = 5, xmax = 45, ymin = -75, ymax = -75) +
annotation_custom(textGrob("'Consolidate land'\npreferable", x = unit(0.8, "npc"),
    y = unit(-0.1, "npc"), just = "left", gp = gpar(fontsize = 10)),
    xmin = 50, xmax = 50, ymin = -75, ymax = -75) +
annotation_custom(linesGrob(x = unit(c(1, 0), "npc"),
    arrow = arrow(length = unit(0.25, "cm"), type = "open"),
    gp = gpar(col = "black")),
    xmin = -5, xmax = -45, ymin = -75, ymax = -75) +
annotation_custom(textGrob("'Maintain farms'\npreferable", x = unit(0.8, "npc"),
    y = unit(-0.1, "npc"), just = "left", gp = gpar(fontsize = 10)),
    xmin = -85, xmax = -85, ymin = -75, ymax = -75)

```

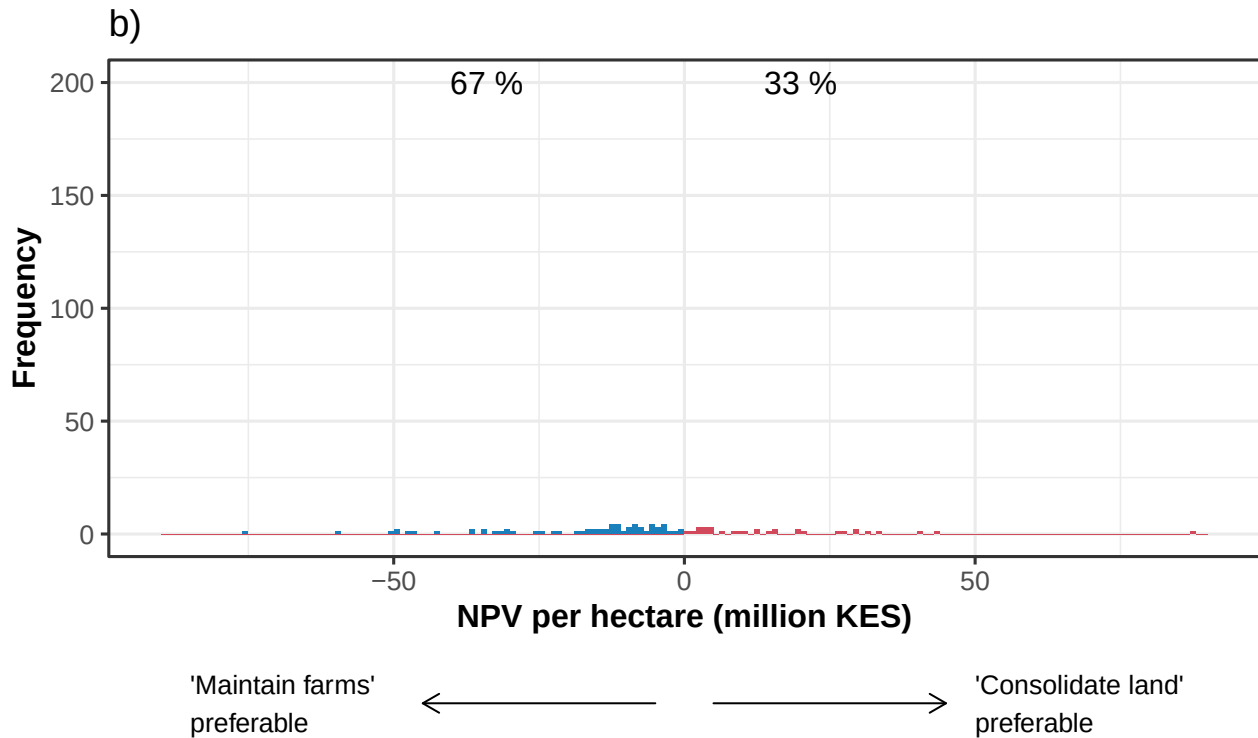


Figure 2: *Distribution of the decision outcomes given as net present value per hectare. Based on 10,000 Monte Carlo model runs.*

Variable importance and Expected Value of Perfect Information

We use variable in projection (VIP) to show important variables in the model and value of information is visualized using the expected value of perfect information (EVPI).

```
pls_result <- plsrmcSimulation(object = mcSimulation_results,
                              resultName = names(mcSimulation_results$y[3]), ncomp = 1)
decisionSupport::plot_pls(
  pls_result, input_table = input_table, threshold = 1,
  pos_color = "#1a80bb", neg_color = "#ea801c", x_axis_name = "VIP score",) +
  scale_y_discrete(labels = \ (y) str_wrap(y, width = 15)) +
  labs(title = "a) ") + #Variable importance
  theme_ltc + theme(legend.position = "bottom",
                    legend.title = element_blank())
```

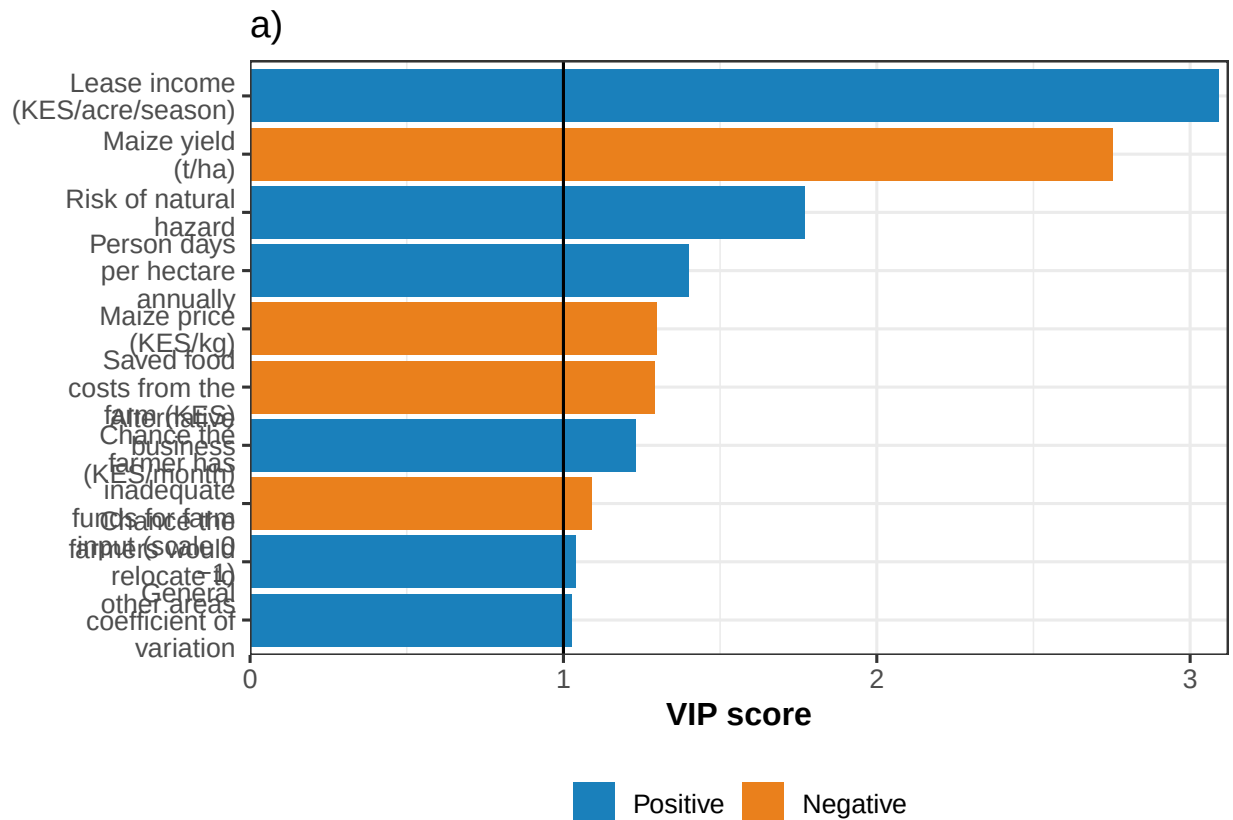
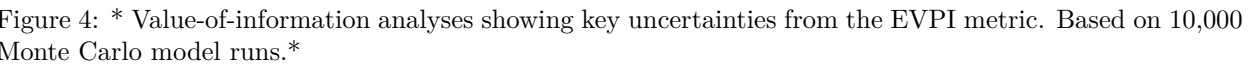


Figure 3: Sensitivity using important variables from PLS regression (VIP scores). Based on 10,000 Monte Carlo model runs.

```
mcSimulation_table <- data.frame(mcSimulation_results$x, mcSimulation_results$y[1:3])
if (!exists("evpi")) {
  evpi <- decisionSupport::multi_EVPI(mc = mcSimulation_table, first_out_var = "benefit_lc_kes")
}
```

```
## [1] "Processing 3 output variables. This can take some time."
```

```
plot_evpi(evpi, decision_vars = "net_benefit_lc_kes",
          input_table = input_table, bar_color = "#1a80bb",
          x_axis_name = "EVPI in million KES",
          new_names = "") +
  scale_y_discrete(labels = \ (y) str_wrap(y, width = 20)) +
  scale_x_continuous(labels = scales::label_number(scale = 1e-6)) +
  labs(title = "b") + #Value of information
  theme_lc
```



Based on two most critical variables in Figure 2 — maize and lease prices — we built scenarios showing how the decision changes as those prices vary.

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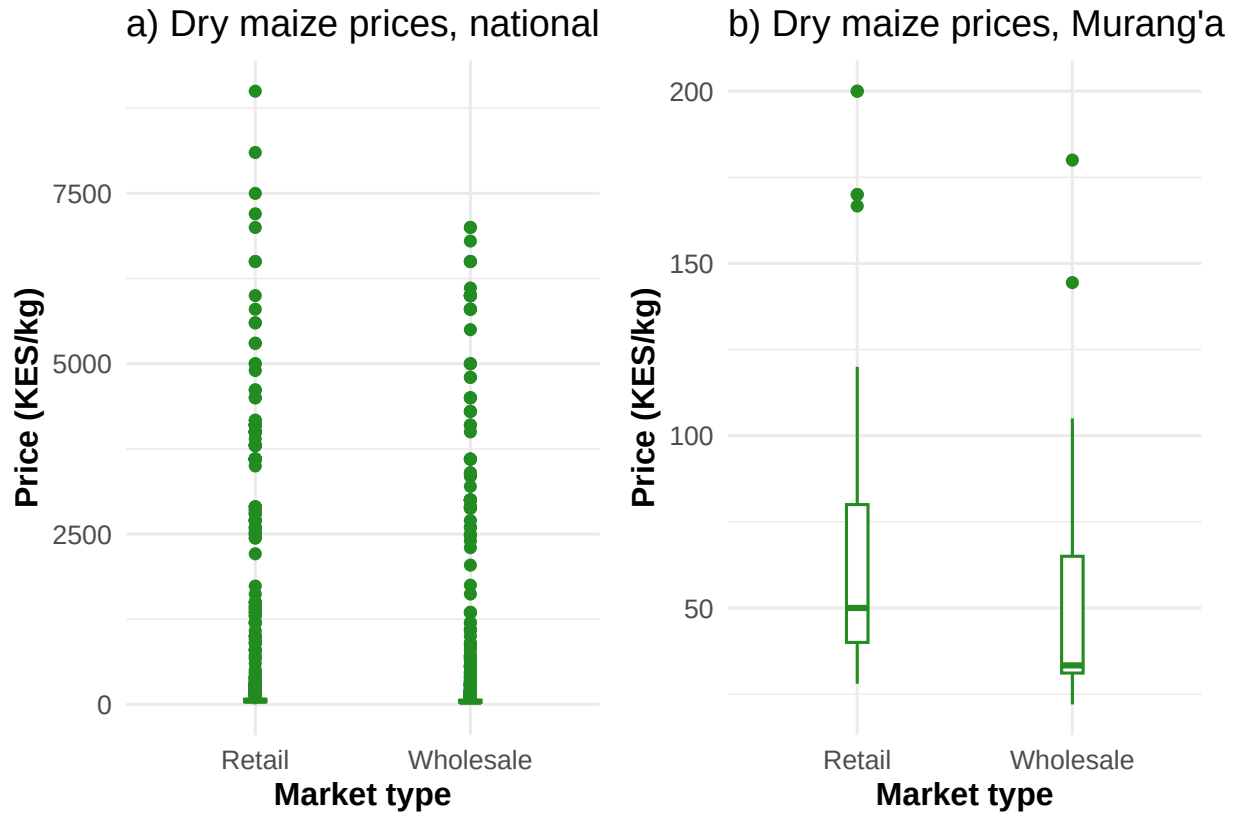
For maize-price bounds we obtained data from daily submissions to the Ministry of Agriculture (<https://amis.co.ke/site/market>). We used national level data and for Murang'a County that was available (2005–2024).

```
# Read every maize-price CSV from the folder and clean the price columns.
maize_files <- list.files("maize_prices/", pattern = "\\*.csv$", full.names = TRUE)

combined_df <- maize_files %>%
  map(\(f) read.csv(f, sep = ";") %>% mutate(Supply.Volume = as.numeric(Supply.Volume))) %>%
  list_rbind() %>%
  mutate(across(c(Wholesale, Retail),
    \(\v) as.numeric(na_if(trimws(gsub("/Kg|,", "", v)), "-"))),
    Date = as.Date(Date))

price_boxplot <- function(df, title) {
  df %>%
    select(Wholesale, Retail) %>%
    pivot_longer(everything(), names_to = "market_type", values_to = "prices") %>%
    ggplot(aes(factor(market_type), prices)) +
    geom_boxplot(width = 0.1, colour = "forestgreen") +
    labs(x = "Market type", y = "Price (KES/kg)", title = title) +
    theme_minimal(base_size = 12) +
    theme(axis.title = element_text(face = "bold"))
}

price_boxplot(combined_df, "a) Dry maize prices, national") +
  price_boxplot(filter(combined_df, County == "Muranga"), "b) Dry maize prices, Murang'a")
```

Maize prices have varied over time.

```
price_timeseries <- function(df, title) {
  df %>%
    mutate(month = as.Date(format(Date, "%Y-%m-01"))) %>%
    group_by(month) %>%
    summarise(Retail = mean(Retail, na.rm = TRUE),
              Wholesale = mean(Wholesale, na.rm = TRUE),
              .groups = "drop") %>%
    pivot_longer(c(Retail, Wholesale), names_to = "Price type", values_to = "price") %>%
    ggplot(aes(month, price, colour = `Price type`)) +
    geom_line(linewidth = 1) +
    labs(x = "Year", y = "Mean price (KES/kg)", title = title, colour = NULL) +
    theme_minimal(base_size = 12) +
    theme(axis.title = element_text(face = "bold"),
          axis.text.x = element_text(angle = 45, hjust = 1))
}

price_timeseries(combined_df, "a) Monthly maize prices over time") +
  price_timeseries(
    filter(combined_df, Date >= as.Date("2021-01-01"),
           Date <= as.Date("2024-12-31")),
    "b) Monthly maize prices, 2021-2024") +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```

The `decisionSupport::scenario_mc` function updates specific variables from a scenario file. Our

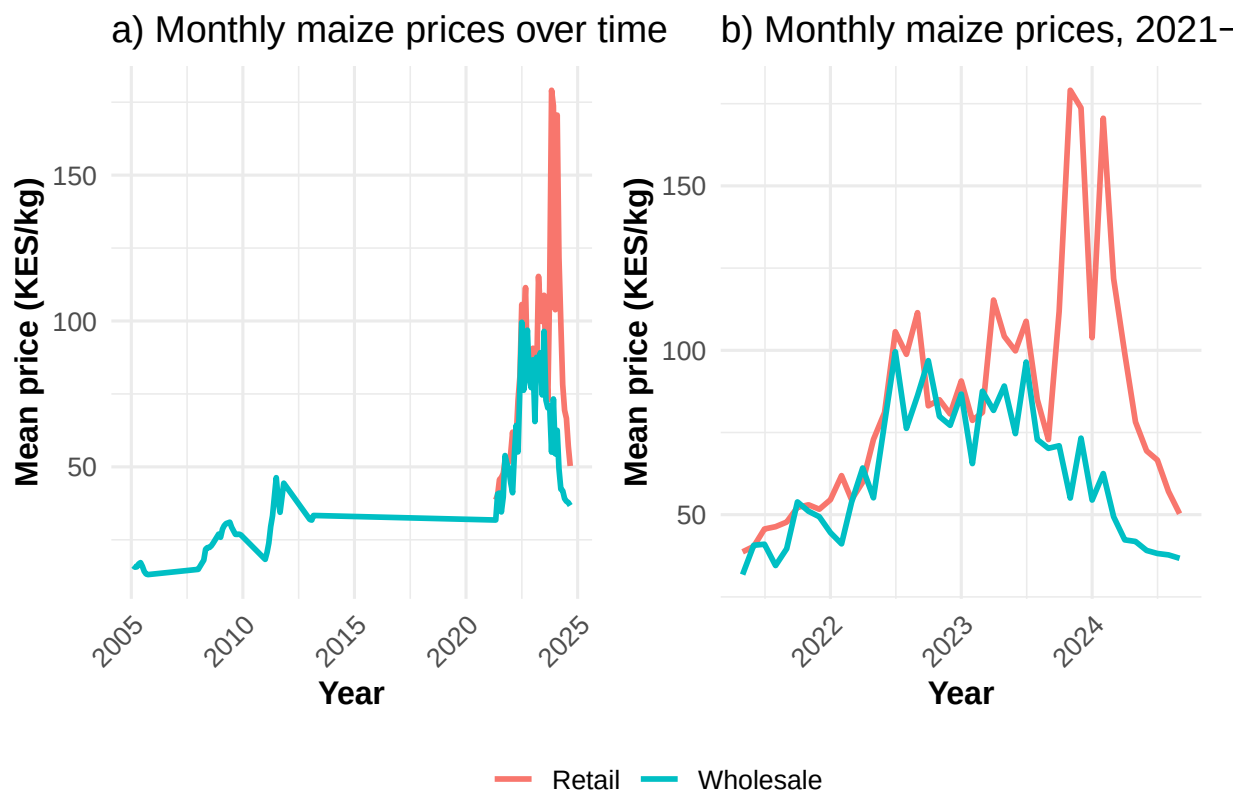


Figure 5: **Figure S2.** Monthly mean maize prices (KES/kg) across the recorded markets: (a) the full series and (b) 2021–2024. Retail and wholesale prices are shown separately.

Variable	param	Scen_1	Scen_2	Scen_3	Scen_4	Scen_5	Scen_6	Scen_7	Scen_8	Scen_9	Scen_10	Scen_11	Scen_12	Scen_13	Scen_14	Scen_15
maize_price_kes_kg	lower	5	86.68	168.34	NA	NA	NA	5	5	5	86.68	86.68	86.68	168.34	168.34	168.34
maize_price_kes_kg	upper	86.67	168.33	250	NA	NA	NA	86.67	86.67	86.67	168.33	168.33	168.33	250	250	250
maize_price_kes_kg	distribution	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm
compensation_income_pn_acre	lower	NA	NA	NA	10000	340001	670001	10000	340001	670001	10000	340001	670001	10000	340001	670001
compensation_income_pn_acre	upper	NA	NA	NA	340000	670000	1000000	340000	670000	1000000	340000	670000	1000000	340000	670000	1000000
compensation_income_pn_acre	distribution	NA	NA	NA	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm	posnorm

market_scenarios.csv updates maize and lease prices.

Table 1. Model input variables updated to generate the market scenarios, chosen from the EVPI results (Figure 2).

```
scenarios <- read.csv("market_scenarios.csv", sep = ";")
if (knitr::is_latex_output()) {
  scenarios %>%
    kbl(booktabs = TRUE, row.names = FALSE) %>%
    kable_styling(latex_options = c("striped", "scale_down"))
} else {
  scenarios %>%
    datatable(extensions = "Buttons", rownames = FALSE,
              options = list(dom = "Blfrtip", buttons = c("copy", "csv", "excel"),
                           scrollX = TRUE, autoWidth = TRUE))
}
```

```
land_consolidation_scenarios <- scenario_mc(
  base_estimate = base_estimate,
  scenarios = scenarios,
  model_function = land_consolidation_function,
  numberOfModelRuns = 1e2,
  functionSyntax = "plainNames")
```

Impact of the market-price scenarios

Table 2 compares the net benefit (the decision) of each scenario against the baseline.

```
# Standardise the combined scenario run to million KES per hectare.
land_consolidation_scenarios$y$net_benefit_lc_per_ha_MKes <-
  land_consolidation_scenarios$y$net_benefit_lc_kes /
  land_consolidation_scenarios$x$ha_per_hh / 1e6

mcSimulation_results$y$net_benefit_lc_per_ha_MKes <-
  mcSimulation_results$y$net_benefit_lc_kes /
  mcSimulation_results$x$ha_per_hh / 1e6
baseline_lc <- mcSimulation_results$y$net_benefit_lc_per_ha_MKes

# Pull each scenario's per-ha net benefit from the single combined run.
scen_keys <- c("low_mp", "med_mp", "high_mp", "low_lp", "med_lp", "high_lp",
              "low_mp_lp", "low_mp_med_lp", "low_mp_high_lp", "med_mp_low_lp",
              "med_mp_lp", "med_mp_high_lp", "high_mp_low_lp", "high_mp_med_lp", "high_mp_lp")
scen_values <- setNames(lapply(seq_along(scen_keys), function(i) {
  land_consolidation_scenarios$y$net_benefit_lc_per_ha_MKes[
    land_consolidation_scenarios$x$Scenario == paste0("Scen_", i)]
}), scen_keys)
list2env(scen_values, envir = environment())
```

```
## <environment: R_GlobalEnv>
```

```
# Display labels, shared by the table and figures.
```

```
scenario_labels <- c(
  baseline_lc = "Baseline",
  low_mp = "Low MP", med_mp = "Medium MP", high_mp = "High MP",
  low_lp = "Low LP", med_lp = "Medium LP", high_lp = "High LP",
  low_mp_lp = "Low MP and LP", low_mp_med_lp = "Low MP and Medium LP",
  low_mp_high_lp = "Low MP and High LP", med_mp_low_lp = "Medium MP and Low LP",
  med_mp_lp = "Medium MP and LP", med_mp_high_lp = "Medium MP and High LP",
  high_mp_low_lp = "High MP and Low LP", high_mp_med_lp = "High MP and Medium LP",
  high_mp_lp = "High MP and LP")

qs <- c(0, 0.05, 0.5, 0.95, 1)
all_series <- c(list(baseline_lc = baseline_lc), scen_values)
scenarios_summary_df <- map_dfr(all_series, \(v) round(quantile(v, qs, na.rm = TRUE), 2)) %>%
  as.data.frame()
colnames(scenarios_summary_df) <- c("Minimum", "5% quantile", "Median", "95% quantile", "Maximum")
scenarios_summary_df <- tibble(`Market price scenario` = scenario_labels[names(all_series)],
  scenarios_summary_df)

if (knitr::is_latex_output()) {
  kable(scenarios_summary_df, format = "latex", booktabs = TRUE, longtable = TRUE,
    caption = "Table 2. Summary statistics of the net benefit (decision) by market-price scenario, ",
    kable_styling(latex_options = c("striped", "repeat_header", "scale_down"), font_size = 8) %>%
    add_header_above(c("Market price scenario" = 1, "Quantiles" = 5), underline = TRUE)
} else {
  kable(scenarios_summary_df, format = "html", caption = "<i><b>Table 2.</b> Summary statistics of the net benefit (decision) by market-price scenario, ",
    kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
    add_header_above(c("Market price scenario" = 1, "Quantiles" = 5), underline = TRUE)
}
```

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

Table 1: Table 2. Summary statistics of the net benefit (decision) by market-price scenario, in million KES per hectare. Based on 10,000 runs.

Market price scenario	Quantiles				
Market price scenario	Minimum	5% quantile	Median	95% quantile	Maximum
Baseline	-1253.83	-89.80	-8.76	17.13	29.74
Low MP	-2646.15	-102.39	2.35	24.25	40.58
Medium MP	-16244.78	-61.02	-5.54	18.59	1352.28
High MP	-492.34	-66.60	-17.35	17.67	28.90
Low LP	-670.49	-94.78	-4.88	22.50	104.48
Medium LP	-1312.83	-13.14	32.06	69.70	90.07
High LP	-1223.61	21.75	65.92	113.27	143.91
Low MP and LP	-4582.45	-74.54	4.89	26.20	41.87
Low MP and Medium LP	-2589.90	-63.53	38.62	70.73	94.12
Low MP and High LP	-1140.34	38.22	72.18	115.10	147.21
Medium MP and Low LP	-196.12	-37.67	-3.02	17.26	30.85
Medium MP and LP	-1916.94	-17.90	27.62	71.98	1622.78
Medium MP and High LP	-210.26	29.56	69.08	101.44	144.80
High MP and Low LP	-535.40	-55.68	-12.25	14.95	30.05

Table 1: Table 2. Summary statistics of the net benefit (decision) by market-price scenario, in million KES per hectare. Based on 10,000 runs. *(continued)*

Market price scenario	Minimum	5% quantile	Median	95% quantile	Maximum
High MP and Medium LP	-2103.38	-17.75	26.73	52.54	97.06
High MP and LP	-3609.32	-10.31	60.34	100.20	147.78

Preference is the likelihood of a positive net benefit (decision) in each scenario. We also examine how each scenario shifts the net benefit relative to the baseline.

```
scenario_order <- c("low_mp", "med_mp", "high_mp",
                  "low_lp", "med_lp", "high_lp",
                  "low_mp_lp", "low_mp_med_lp", "low_mp_high_lp",
                  "med_mp_low_lp", "med_mp_lp", "med_mp_high_lp",
                  "high_mp_low_lp", "high_mp_med_lp", "high_mp_lp")

scenario_labels <- c(
  low_mp      = "Low maize price",
  med_mp      = "Medium maize price",
  high_mp     = "High maize price",
  low_lp      = "Low lease price",
  med_lp      = "Medium lease price",
  high_lp     = "High lease price",
  low_mp_lp   = "Low maize price and lease price",
  low_mp_med_lp = "Low maize price and medium lease price",
  low_mp_high_lp = "Low maize price and high lease price",
  med_mp_low_lp = "Medium maize price and low lease price",
  med_mp_lp   = "Medium maize price and lease price",
  med_mp_high_lp = "Medium maize price and high lease price",
  high_mp_low_lp = "High maize price and low lease price",
  high_mp_med_lp = "High maize price and medium lease price",
  high_mp_lp   = "High maize price and lease price")

change_summary <- as_tibble(scen_values) %>%
  mutate(across(everything(), \(v) v - baseline_lc)) %>% # per-run change vs baseline
  pivot_longer(everything(), names_to = "scenario", values_to = "change") %>%
  group_by(scenario) %>%
  summarise(p05 = quantile(change, 0.05, na.rm = TRUE),
            median = median(change, na.rm = TRUE),
            p95 = quantile(change, 0.95, na.rm = TRUE),
            .groups = "drop") %>%
  mutate(label = scenario_labels[scenario],
         label = factor(label, levels = rev(scenario_labels[scenario_order])),
         direction = ifelse(median >= 0, "More attractive", "Less attractive"))

ggplot(change_summary,
       aes(x = median, y = label, colour = direction)) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey40") +
  geom_pointrange(aes(xmin = p05, xmax = p95), linewidth = 0.7, fatten = 3) +
  scale_colour_manual(values = c("More attractive" = "#1a80bb",
                                "Less attractive" = "#d1495b"), name = NULL) +
  labs(x = "Change in NPV per hectare vs baseline (million KES)",
       y = NULL) + #,
```

```
# title = "How each market-price scenario shifts the decision for land consolidation") +
theme_minimal(base_size = 12) +
theme(legend.position = "top",
      axis.title = element_text(face = "bold"))
```

```
## Warning: The 'fatten' argument of 'geom_pointrange()' is deprecated as of ggplot2 4.0.0.
## i Please use the 'size' aesthetic instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

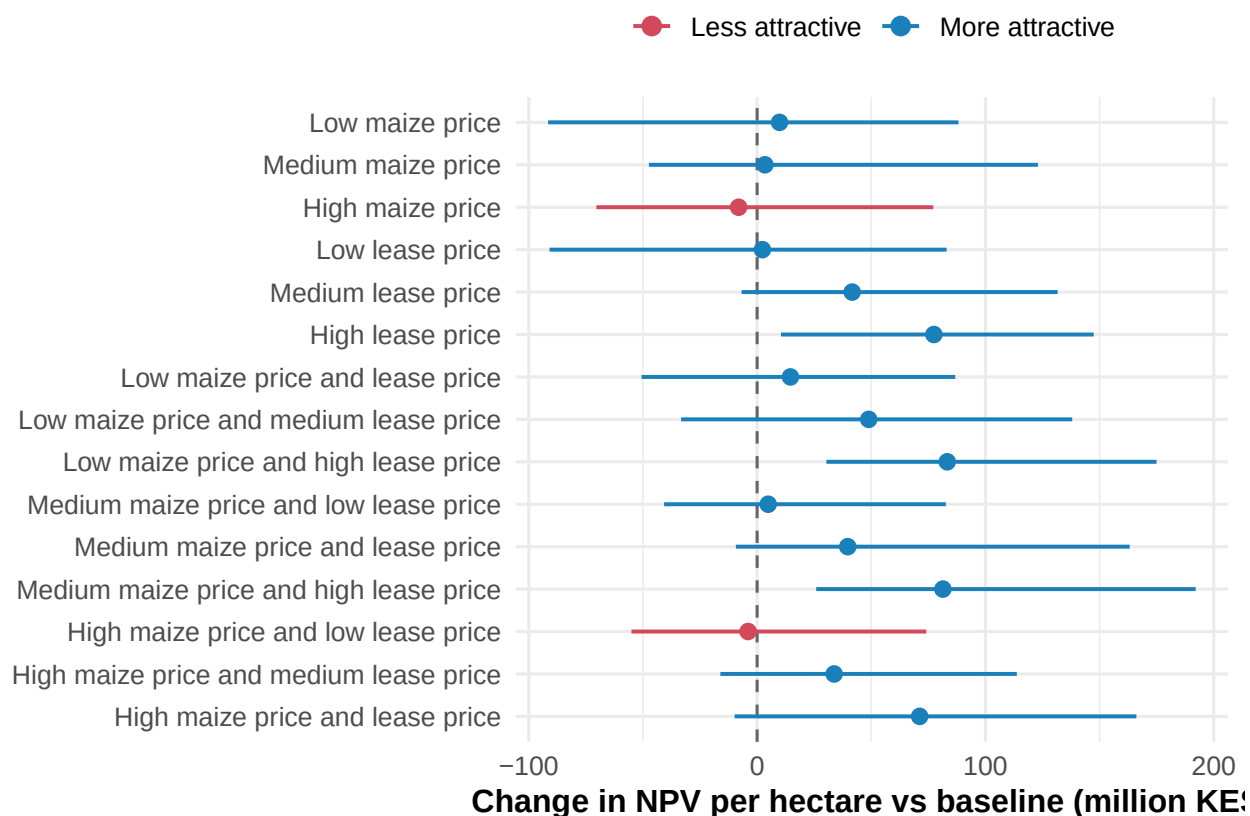


Figure 6: Change in the decision (scenario minus baseline), per hectare. Point = median change; bar = 90% interval. Colour shows whether the scenario makes consolidation more or less attractive. Based on 10,000 runs.

References

- Luedeling, Eike, Lutz Goehring, Katja Schiffrers, Cory Whitney, and Eduardo Fernandez. 2025. *decisionSupport: Quantitative Support of Decision Making Under Uncertainty*. <http://www.worldagroforestry.org/>.
- Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>.
- Wickham, Hadley. 2023. *Tidyverse: Easily Install and Load the Tidyverse*. <https://tidyverse.tidyverse.org>.